Simulating Climate Risk Migration

The Shadow Banking Spillover Effect

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October 2026

Outline

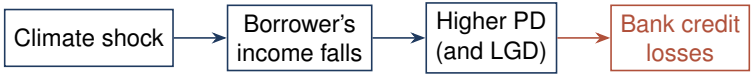
1. Background
2. Model design
3. Mechanism
4. Worked example
5. Results
6. Conclusion

The one question

When a bank prices climate risk *correctly*, does the risk disappear—or does it move somewhere else?

Why should a bank care about climate?

A bank does not lend against the weather; it lends to *borrowers* and cares whether they keep paying.



	Physical risk	**Transition risk**
Source	Floods, drought	Carbon tax, coal phase-out
Hits	Farms, coasts	Power, steel, cement
Channel	Income, collateral	Profit, stranded assets

Expected loss

$EL = PD \times LGD \times EAD$

A flood hits a loan of 500: PD 3% $\rightarrow$ 15% at LGD 60% takes EL from 9 to 45.

PD: probability of default; LGD: loss given default; EAD: exposure at default.

What happens to a borrower the bank turns away?

What we already know (India)

- After local floods, bank branches raise rates (Das, Majilla & Saurav).
- Non-banks (NBFCs) expand lending in the same areas (Mathur & Sarma).

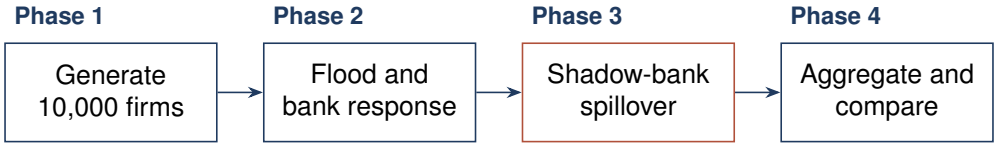
NBFC: lends like a bank, takes no ordinary deposits, lighter regulation. With similar lenders it forms the "shadow banking" sector.

Two testable implications

1. The bank's remaining loan book looks *safer* after the shock.
2. The firms pushed out default at a *far higher* rate.

> **Hypothesis.** Strict climate regulation of banks does not eliminate systemic climate risk; it moves it into the opaque shadow-banking sector.

A synthetic economy, built in four phases



Why a simulation?

No public dataset links, for the same firm, its flood exposure, its bank's decision and its later NBFC borrowing.

Why synthetic data?

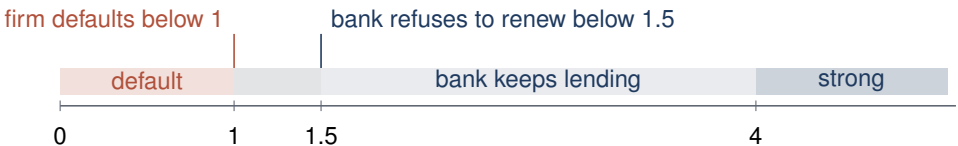
It isolates the mechanism. Firms are calibrated to published RBI data on firm health, so the starting point is not arbitrary.

> This is a **proof of mechanism**, not a forecast for the Indian economy.

Repayment capacity in one number: the ICR

$$ICR = \frac{\text{EBITDA}}{\text{interest expense}}$$

"How many times does operating profit cover the interest bill?"



- A flood lowers the numerator (income). Lenders' response raises the denominator (interest).
- ICR below 1.5 is widely read as acute financial stress.

Phase 1: build 10,000 firms, anchored to RBI data

Each firm holds one bank loan at 8.5% before the flood.

- Revenue: normal(500, 150), clipped to 100–2000
- EBITDA margin: uniform, 10–30%
- Debt = multiple of revenue: 0.1–0.6× (50% of firms), 0.6–2.5× (25%), 2.5–4.5× (25%)
- Flood vulnerability v: integer 1–100, independent of finances

Is the baseline realistic?

Baseline ICR	Model	RBI
Strong (> 4)	43.8%	47.6%
Moderate (1–4)	29.0%	30.5%
Vulnerable (≤ 1)	27.2%	21.9%

Close to the RBI pattern, with slightly more vulnerable firms.

RBI: listed non-financial firms, H1 2023–24 (report of 28 Dec 2023).

Phase 2: the flood hits income; banks reprice and ration

1. **Shock.** Revenue falls by 35% $\times$ $v/100$. A firm with $v = 100$ loses 35%; margins are held fixed, so EBITDA falls in proportion.
2. **Reprice.** Banks raise the rate from 8.5% to 9.0% for firms with $v > 50$.
3. **Ration.** Banks renew a loan only if the stressed ICR is at least 1.5.

Outcome in the simulation

Aggregate EBITDA	–17.6%
Loans renewed	5,896
Loans refused	4,104 (41%)

About 90% of refused firms already had an ICR below 1.5 before the flood: the shock tips weak firms over the edge.

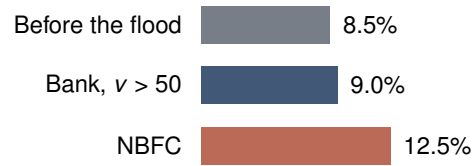
> Banks here do exactly what regulators want: they refuse firms that have become too risky.

Phase 3: refused firms turn to NBFCs

- **Full absorption.** NBFCs lend to every refused firm.
- **Risk premium.** They add a 400 bps premium: 12.5%, up from 8.5%.
- **Default.** A firm defaults if its final ICR is below 1: profit cannot even pay the interest.

Default here means financial distress (interest not covered), not a legal bankruptcy.

Interest rate each firm pays



> The 400 bps premium is illustrative, and it is the most influential parameter of this phase.

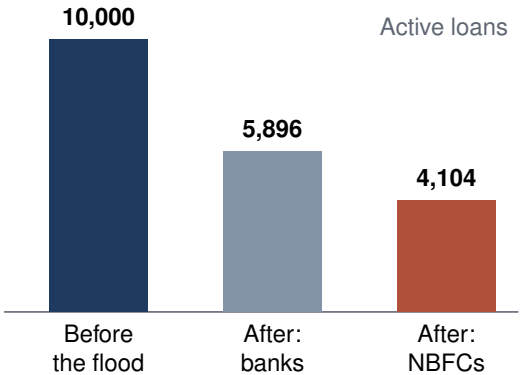
One firm, end to end

A firm with EBITDA of 100, debt of 600 and flood vulnerability $v = 80$ (revenue and EBITDA fall by $0.35 \times 0.80 = 28\%$). Interest = debt $\times$ rate; ICR = EBITDA / interest.

Stage	EBITDA	Rate	Interest	ICR	Outcome
Before the flood	100	8.5%	51	1.96	healthy
After flood, bank rate	72	9.0%	54	1.33	refused (below 1.5)
After flood, NBFC rate	72	12.5%	75	0.96	default (below 1)

> The firm is the same and its debt is the same. Only the shock and the lender changed, and the firm moved from healthy to distressed.

Result 1: credit migrates, and the bank book looks safer

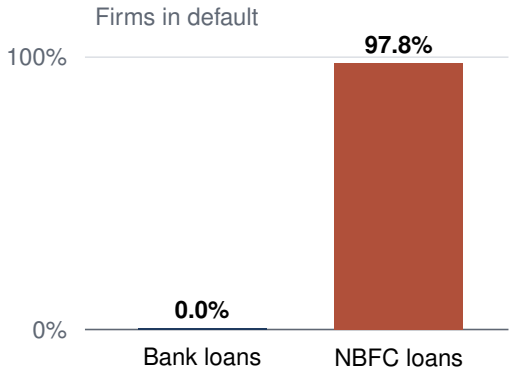


41% of loans move to NBFCs, but they carry **81%** of all debt: the firms that move are the most heavily indebted.

Among firms that stay with banks, the share with a strong pre-shock ICR rises from **44%** to **74%**.

A stress test that looks only at banks would see a stronger portfolio after the flood.

Result 2: the migrated risk is fragile



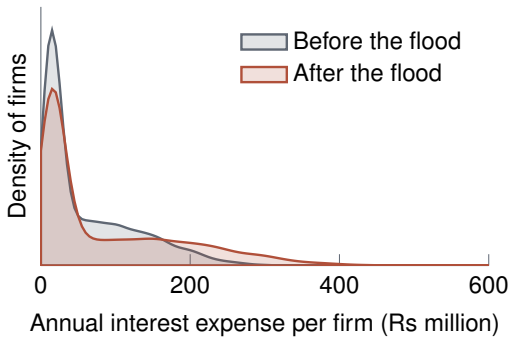
0 of 5,896 bank borrowers default.

4,013 of 4,104 NBFC borrowers default (97.8%); only 91 survive.

All defaults sit in a portfolio that bank-level statistics do not show.

Partly by construction: these firms were already weak and the premium is large. See the limitations.

Result 3: the economy-wide interest burden rises



Total interest paid by the economy rises by **39%**.

Two drivers:

- a higher bank rate for exposed firms
- heavily indebted firms moving to a lender that charges 12.5%

The right tail of large interest bills becomes thicker.

Conclusion: what we learned, and how far to trust it

What we learned

1. Correct pricing of climate risk by banks does not remove the risk from the economy.
2. The risk migrates to lighter-regulated lenders, where it is harder to see.
3. Bank-only stress tests can look reassuring while systemic risk grows.

How far to trust it

- Synthetic firms: a proof of mechanism, not a forecast.
- One-off, linear shock; no collateral or recovery.
- Full NBFC absorption and the 400 bps premium are assumed.
- Default means ICR below 1, not bankruptcy.

> **Next step:** vary the premium and the 1.5 cutoff to test how sensitive the results are, then use real lender-level data.

References

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Thank you

Questions and discussion